REQUIREMENT ANALYSIS DOCUMENT

TITLE: Script-Controlled Access Control List (ACL) to Restrict Record Access Based on Field Value in ServiceNow

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INTRODUCTION:

ServiceNow is a cloud-based platform used by organizations to manage workflows, applications, data, and business processes. Since applications may contain sensitive information, controlling access to records is an important requirement.

ServiceNow provides **Access Control Lists (ACLs)** to determine whether a user is permitted to perform a particular operation on a particular record.

The major operations controlled through ACLs include:

- Read
- Write
- Create
- Delete

In this project, ACLs are implemented on a student/institution details table. Different roles are assigned for different operations so that users receive only the required permissions.

The project also demonstrates the use of a **scripted ACL**. The script checks the user's role and determines whether access should be allowed. A data condition is used to restrict the Read operation based on the **Branch** field

Functional Requirements

FR1 – User Creation

The system shall allow the administrator to create a user named:

EEE User

FR2 – Role Creation

The system shall provide custom roles:

- `bb1`
- `bb2`
- `bb3`
- `bb4`

In the implemented ACL configuration, `bb1`, `bb2`, and `bb4` are used.

FR3 – Table Creation

The system shall contain an institution/student details table.

FR4 – Student Roll Number

The table shall contain a **Student Roll Number** field configured as an Auto Number.

FR5 – Student Name

The table shall contain a field for storing the student's name.

FR6 – Faculty Name

The table shall contain a field for storing the faculty name.

FR7 – Branch

The table shall contain a Branch field used for identifying the student's branch.

FR8 – Phone Number

The table shall contain a Phone Number field.

FR9 – Description

The table shall contain a Description field for additional information.

FR10 – Read Access

The system shall allow Read access through the `bb1` role.

FR11 – Branch-Based Read Restriction

The Read access shall include the condition:

Branch is EEE

FR12 – Write Access

The system shall allow Write access to users having the `bb2` role.

FR13 – Create Access

The system shall allow Create access to users having the `bb4` role.

FR14 – Delete Access

The system shall allow Delete access to users having the `bb4` role.

USER REQUIREMENTS

The main user in the project is:

EEE User

The user's access is controlled using ServiceNow roles and ACLs.

The administrator is responsible for:

- Creating users.
- Creating roles.
- Creating the table.
- Configuring ACLs.
- Assigning roles.
- Testing access permissions.

ROLE REQUIREMENT

The following custom roles are created:

bb1

Used for:

Read Access

bb2

Used for:

Write Access

bb3

Created as a custom role but not used in the current ACL configuration.

bb4

Used for:

- Create Access
- Delete Access

TABLE REQUIREMENT

The project uses an institution/student details table.

Table Fields

Field Name	Purpose
Student Roll Number	Unique student identification number
Student Name	Stores student name
Faculty Name	Stores faculty information
Branch	Identifies the student's branch
Phone Number	Stores contact information
Description	Stores additional details

The **Student Roll Number** is configured as an **Auto Number** so that a unique number can be automatically generated for each record.

ACCESS CONTROL REQUIREMENTS

The system uses ACLs to control access to the table.

ACL Matrix

ACL	Operation	Required Role	Condition	Script
ACL 1	Read	`bb1`	Branch = EEE	Yes
ACL 2	Write	`bb2`	None	No

ACL 3	Create	`bb4`	None	No
ACL 4	Delete	`bb4`	None	No

This ACL matrix defines the access permissions required by the project.

SCRIPTED READ ACL REQUIREMENT

The Read ACL is configured as an Advanced ACL so that a script can be used.

The purpose of the script is to check the logged-in user's role.

The script used is:

```
(function () {
    // Allow admin users full access
    if (gs.hasRole('admin')) {
        return true;
    }

    // Allow users with bb1 role
    if (gs.hasRole('bb1')) {
        return true;
    }

    // Deny access for all others
    return false;
})();
```

The script provides role-based access control.

The **Data Condition** is configured separately as:

Branch is EEE

Therefore, the Read ACL combines the role requirement and branch-based restriction.

WRITE ACL REQUIREMENT

A separate Write ACL is created for the table.

Configuration

- **Type:** Record
- **Operation:** Write
- **Name:** `u_institution_details`
- **Active:** True

- **Requires Role:** `bb2`
- **Advanced:** Not enabled
- **Script:** Not required

The purpose of this ACL is to ensure that only users with the `bb2` role can modify the permitted records.

CREATE ACL REQUIREMENT

A Create ACL is created to control the creation of new records.

Configuration

- **Type:** Record
- **Operation:** Create
- **Name:** `u_institution_details`
- **Active:** True
- **Requires Role:** `bb4`
- **Advanced:** Not enabled
- **Script:** Not required

Only users having the `bb4` role are granted Create permission through this ACL.

DELETE ACL REQUIREMENT

A Delete ACL is created to control the deletion of records.

Configuration

- **Type:** Record
- **Operation:** Delete
- **Name:** `u_institution_details`
- **Active:** True
- **Requires Role:** `bb4`
- **Advanced:** Not enabled
- **Script:** Not required

This prevents users without the required role from deleting records through this ACL.

CONCLUSION

The project successfully demonstrates the implementation of **Access Control Lists in ServiceNow** to control access to student/institution records.

A student details table was created with fields including Student Roll Number, Student Name, Faculty Name, Branch, Phone Number, and Description. Custom roles were created and used to control different operations.

The **Read ACL** uses the `bb1` role together with a **Branch = EEE** condition and a script for role-based control. The **Write ACL** uses the `bb2` role without scripting. The **Create** and **Delete** ACLs use the `bb4` role without scripting.

Overall, the project demonstrates how ServiceNow can provide **role-based and field-based record security** using ACLs. By controlling Read, Write, Create, and Delete operations separately, the system helps protect student information and ensures that users receive only the access required for their responsibilities.